



Conditional Generation from Unconditional Generative Models

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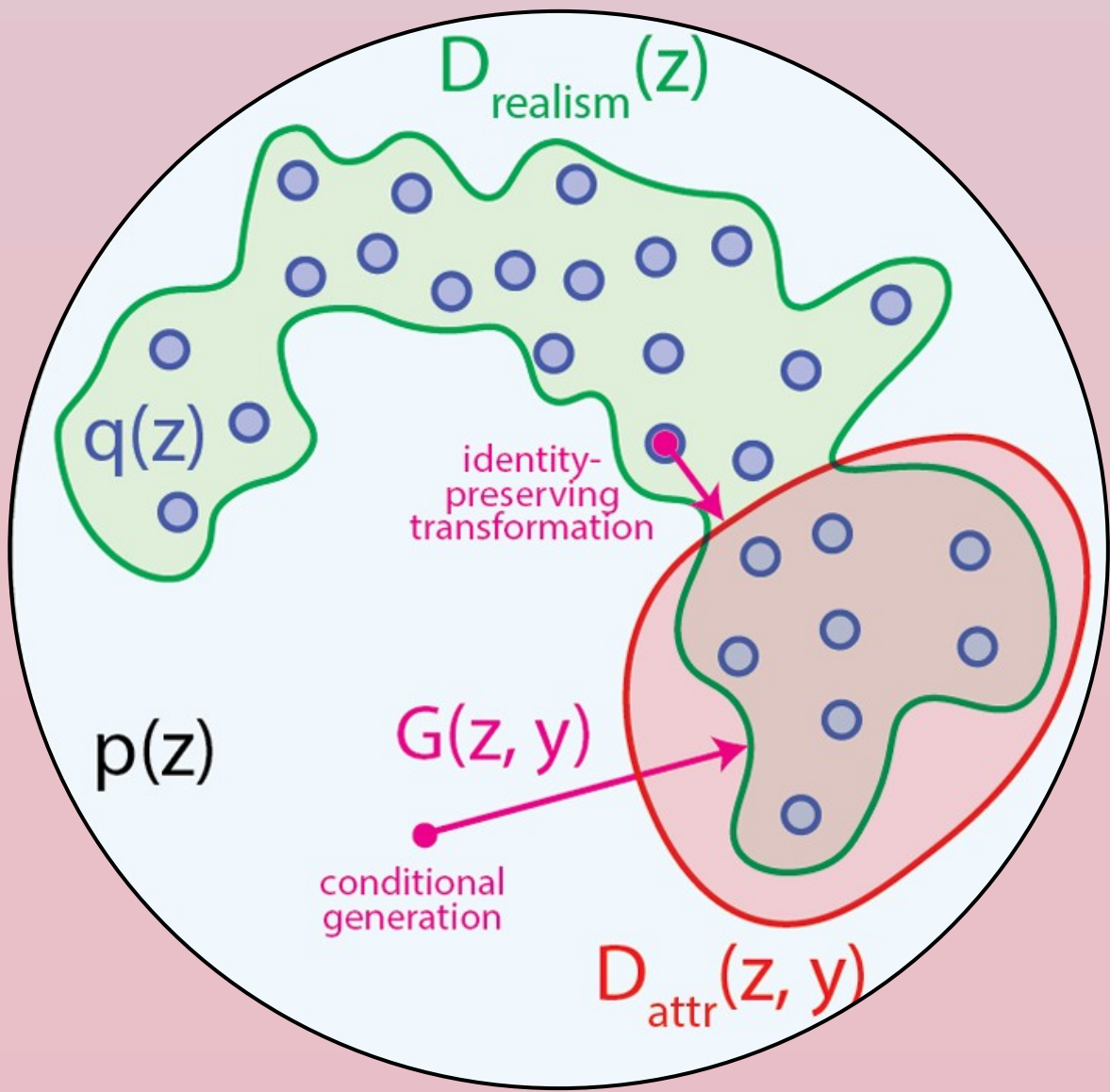
PROBLEM STATEMENT



- Save compute and time
- Understanding the latent space structure of generative models
- Conditional Generation without retraining
- Leverage Existing unconditional models
- Preservation of identity in transformation

Approach

- $p(z)$: Prior Distribution
- y : Attributes of data
- $G(z, y)$: Actor function
- $D_{realism}(z)$: Realism Constraint
- $D_{attr}(z, y)$: Critiq function



Deep generative neural networks can create complex data like images in two ways: by simply generating random outputs (unconditional) or by letting you control what gets generated (conditional). Generally, adding new controls requires retraining the model, which can be very costly. In our work, we show a way to add these controls without having to retrain the model. Instead, we learn patterns in the model's hidden space that mark areas where the generated data has the features we want. Then, we use an actor network to pick points in these areas. By combining rules for specific features with a general rule that makes sure outputs look like real data, we can produce realistic, controlled images from a model that was originally trained without those conditions. We also try to preserve identity while navigating the latent space

TIMELINE

- 1) Learning GANs, VAEs, NST and Style Gan 2
- 2) Replicated the paper : 'Latent Constraints' and thoroughly understanding concepts
- 3) Applied the learnings on style gan 2 for conditional generation

VAE - CelebA

Methodology:

Base Model (VAE):

Trained to capture the underlying data distribution with an emphasis on reconstruction quality. The trained VAE establishes a semantically structured latent space that serves as the foundation for subsequent conditional transformations.

Latent Actor-Critic Pair: The actor network (a GAN-like model) is trained in latent space to shift latent encodings such that desired attributes are emphasized. The critic network learns to evaluate the realism of these modifications in latent space, ensuring that shifts do not compromise the overall image fidelity. A distance penalty is applied to minimize unnecessary deviations, thus **Results:** preserving the original identity of the subject.

VAE Training Time: 64,570 seconds

Actor-Critic Pair Training Time: 11,167 seconds

Attribute Conditioning: Targeted modification for 10 prominent attributes (e.g., hair color, facial hair, eyeglasses, etc.).

The actor-critic framework successfully altered only certain features (such as the addition of glasses or changes in hair style/color) while keeping the facial identity preserved.



StyleGan2

Methodology

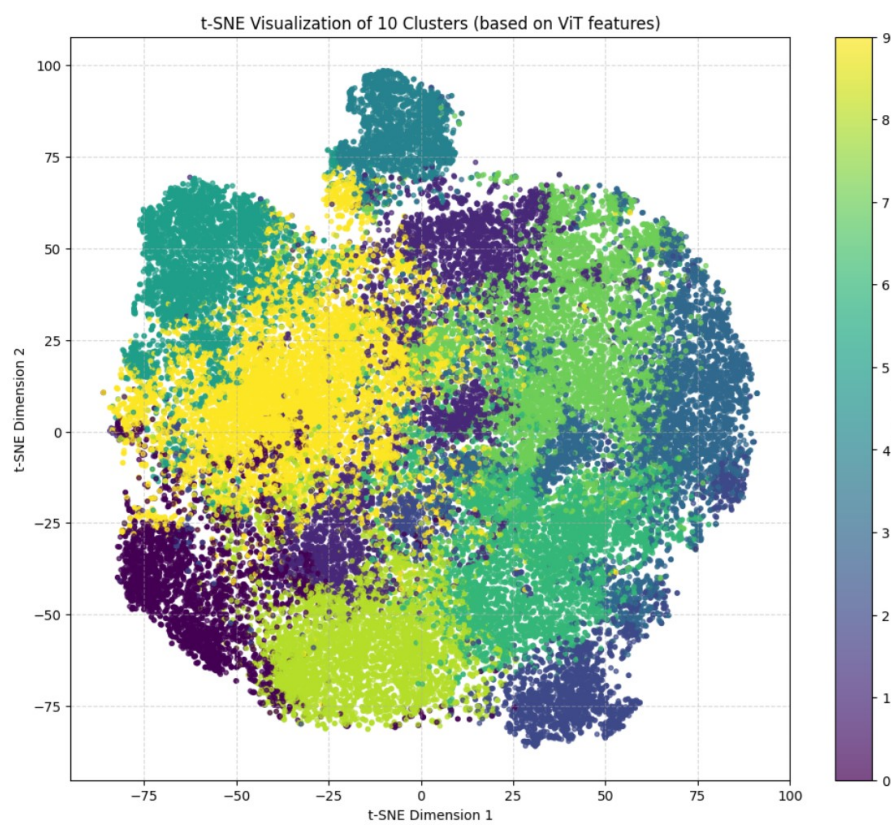
Image Generation: Use a pre-trained StyleGAN2 model (e.g. Car Config E) to generate an initial dataset of 10,000 images.

Feature Extraction: Apply a Vision Transformer (ViT) to extract high-level visual features from the generated images.

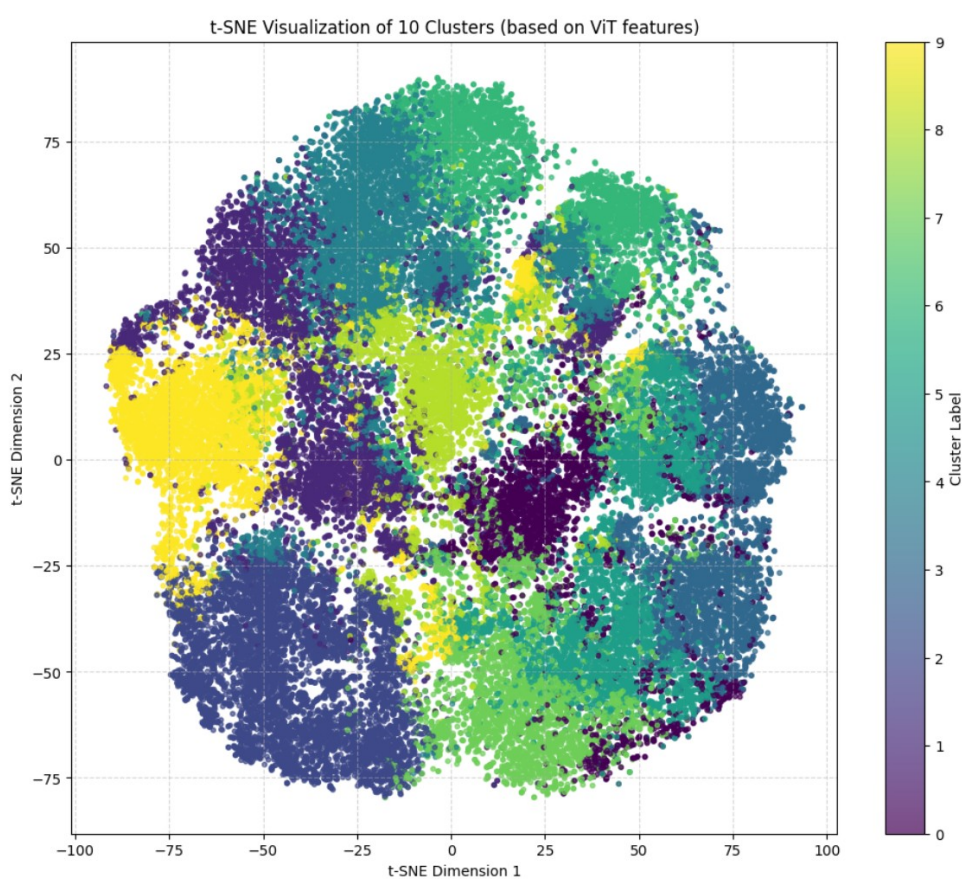
Clustering via K-means: Perform K-means clustering on the extracted features to group similar types of cars. Each cluster ideally represents a distinct car type, effectively partitioning the latent space into meaningful subregions.

Latent GAN Training: Train a separate latent generator on each identified cluster or a single GAN on overall latent space with appending attributes to produce latent codes that correspond to the **Car Config E** type.

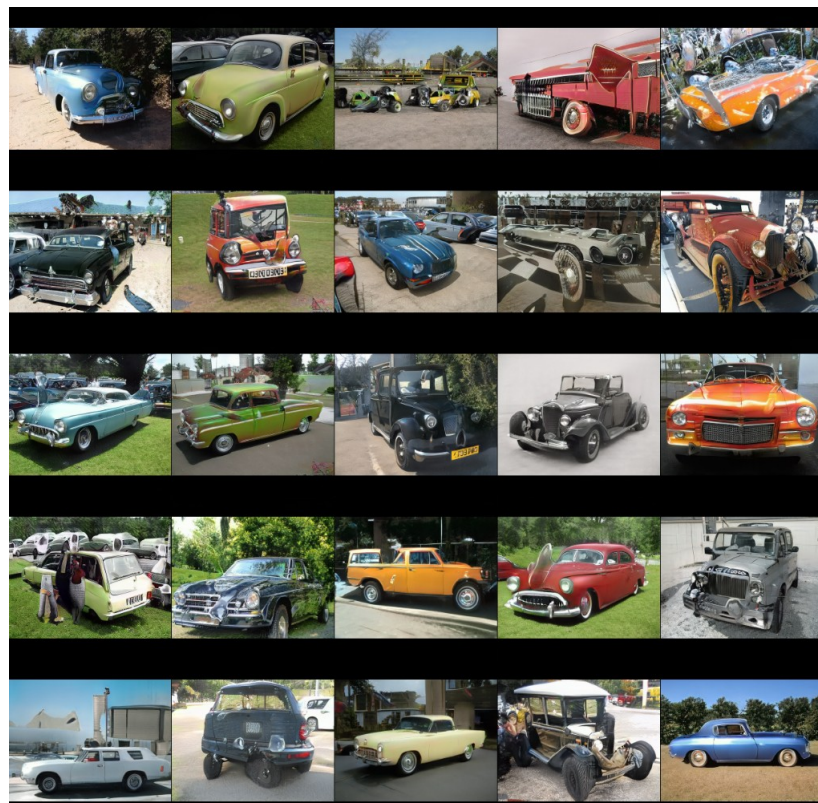
dataset:



FFHQ Dataset:



Samples from one of the cluster



Conditional Regeneration



Samples from one of the cluster



Conditional Regeneration



APPLICATIONS & FUTURE WORK

- **Applications:** There are many unconditional models and using this approach we can directly leverage them without retraining
- **Limitations :** For different architectures and datasets we need to use relevant methods to navigate latent space.
- **Next Steps in Research**
 1. Extend the concept for clip models, LLMs, Energy Models and more ...
 2. Build a library that would integrate over any sort of model and provides utility to leverage the latent space

RÉFÉRENCES

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